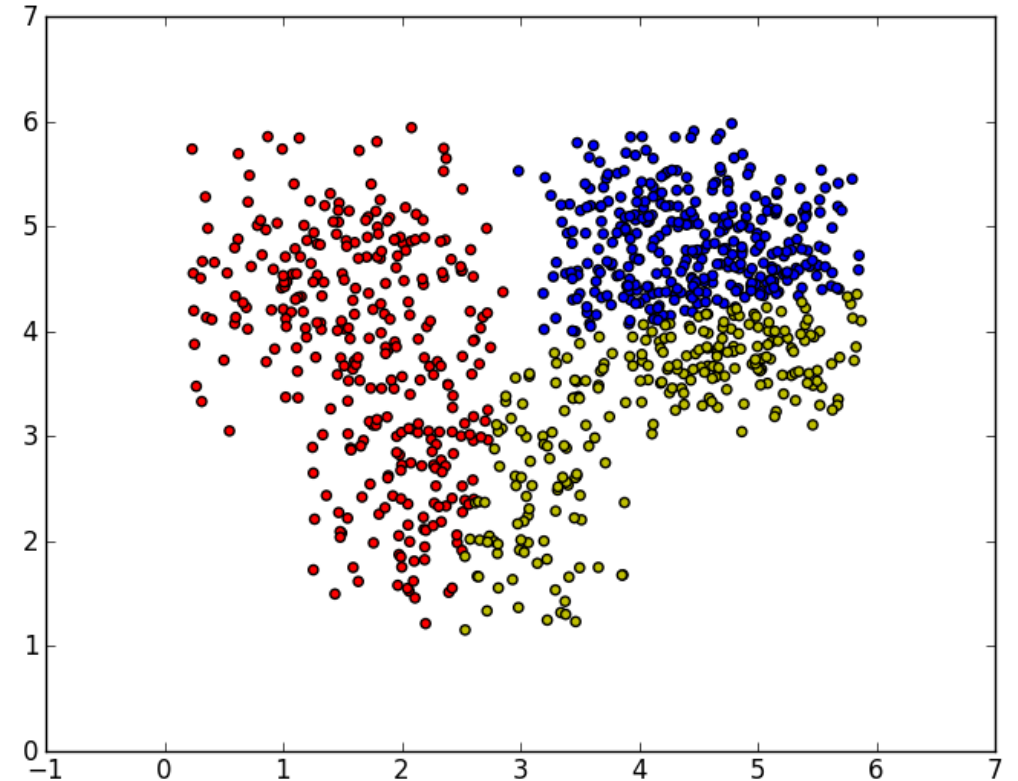


Cluster Analysis

- ❑ Unsupervised learning (i.e., Class label is unknown)
- ❑ Group data to form new categories (i.e., clusters), e.g., cluster houses to find distribution patterns
- ❑ Principle: Maximizing intra-class similarity & minimizing interclass similarity
- ❑ Many methods and applications



Deep Learning

- ❑ Deep learning: A fast expanding dynamic frontier in machine learning
- ❑ Deep learning has developed various *neural network architectures*
 - ❑ Feed-forward neural networks
 - ❑ Convolutional neural networks
 - ❑ Recurrent neural networks
 - ❑ Graph neural networks
 - ❑ Transformer
- ❑ Deep learning has broad applications in computer vision, natural language processing, machine translation, social network analysis, and so on
- ❑ Deep learning has been reshaping a variety of data mining tasks
 - ❑ Ex. classification, clustering, outlier detection, and reinforcement learning

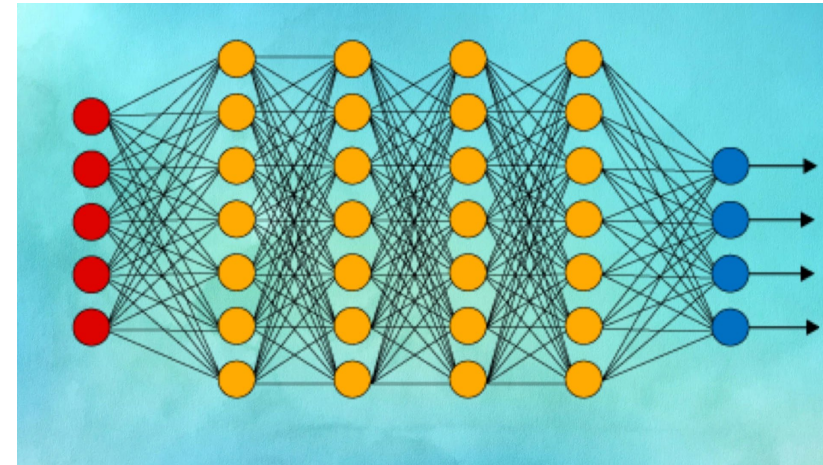


Image Source:
<https://bdtechtalks.com/2021/01/28/deep-learning-explainer/>

Outlier Analysis

- ❑ Outlier analysis
 - ❑ Outlier: A data object that does not comply with the general behavior of the data
 - ❑ Noise or exception?—One person's garbage could be another person's treasure
 - ❑ Methods: by product of clustering or regression analysis, ...
 - ❑ Useful in fraud detection, rare events analysis



Image Source:
<https://www.mathworks.com/matlabcentral/mlc-downloads/downloads/submissions/34795/versions/7/screenshot.png>

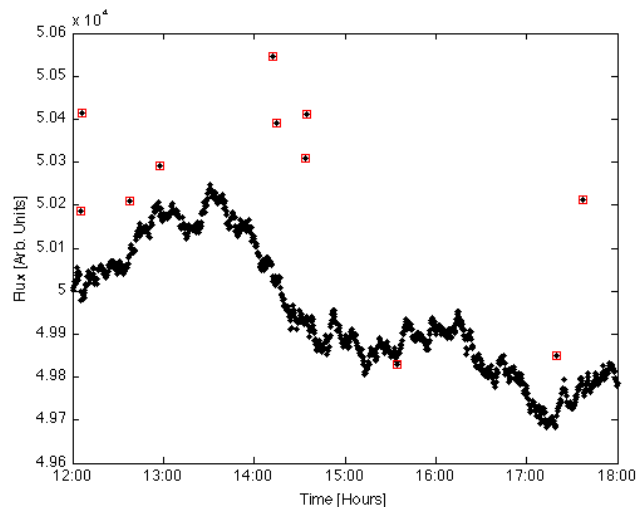


Image Source:
<http://photos1.blogger.com/x/blogger/5682/4111/1600/485624/Multivariate%20Outlier%20Example.png>

